

Agrim Sigdel

Full-Stack Developer | AI Enthusiast

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PROFESSIONAL SUMMARY

Versatile Full-Stack Developer with a strong foundation in modern AI, engineering production-ready applications using React/TypeScript and Python/Django. Adept at building scalable REST APIs, optimizing relational databases, and implementing clean architecture principles across web, mobile, and open-source platforms.

TECHNICAL SKILLS

- **Frontend:** React.js, React Native, JavaScript, TypeScript, Next.js, HTML, CSS, Tailwind CSS
- **Backend:** Python, Django, Django REST Framework (DRF), REST API Design
- **Database:** PostgreSQL, MySQL
- **Tools & DevOps:** Git, GitHub, GitLab, Docker, CI/CD Pipelines, Server Configuration, Postman
- **Specialized:** Computer Vision (YOLO, RT-DETR), NLP/Transformers, Open-Source Development, Creative Leadership

WORK EXPERIENCE

Software Developer (Full-Stack)

Apr 2026 – Present

Contractor Ops AI | Remote

- Engineer a high-performance mobile application using React Native and TypeScript, ensuring strict adherence to component state-management best practices.
- Refactor and optimize backend REST APIs to ensure seamless compatibility, efficient data serialization, and low-latency payloads between server-side systems and client applications.
- Debug and resolve integration bottlenecks across the full stack, improving database query handling and streamlining API endpoints for real-time application features.

Full-Stack Developer

Oct 2025 – Present

Kingsoft Tech | On-site, Kathmandu

- Spearheaded full-stack development, optimization, and deployment for Feriaau and its corresponding multi-tenant CRM platform (crm.feriaau.com).
- Owned the complete development lifecycle, engineering modular Python/Django backend services, designing robust REST APIs, and building dynamic React.js/TypeScript user interfaces.
- Configured secure production server environments and established automated CI/CD deployment pipelines, ensuring high availability, streamlined version control, and seamless code delivery.
- Optimized relational database schemas and queries in PostgreSQL/MySQL to maximize application speed and multi-tenant isolation.

Creative Lead

May 2024 – Sep 2024

Prime College Graduation Committee | Kathmandu

- Led creative direction and brand guidelines for 625+ attendees, coordinating event planning and execution.
- Managed cross-functional teams for visual and marketing materials, practicing agile task delegation.

RESEARCH & PUBLICATIONS

Lead Researcher & First Author

2024 – 2025

Publication: Sigdel, A., Thapa, S., Thapa Magar, B., & Gajurel, R.K. "Cascaded AI Task Decomposition: A Generalizable Framework for Multi-Stage Computer Vision Pipelines Validated on Automated Number Plate Recognition for Nepali License Plates." (Submitted for peer review)

- Formalized the **Cascaded AI Task Decomposition (CATD)** framework as a generalizable design pattern for multi-stage computer vision pipelines with applications beyond ANPR.
- Achieved 97.9% mAP₅₀ plate localization and 98.9% mAP₅₀ character recognition using custom-trained YOLOv11n and RT-DETR Base models on 61-class bi-scriptural Nepali plates.
- Developed a two-dimensional cross-domain training strategy combining 1,380 Nepali traditional plates with 1,127 Indian embossed plates to enable compositional zero-shot transfer (+24.5 pp improvement).
- Built a three-stage cascaded pipeline (YOLOv8n → YOLOv11n → RT-DETR) achieving 90.4% end-to-end precision with 61% – 73% compute reduction through intelligent vehicle gating.

- Integrated the system as the computer vision backbone of a smart parking platform handling automated billing and real-time traffic analytics.

PROJECTS & OPEN SOURCE

easy-3dkit (NPM Package & Documentation) NPM Link | Live Docs

- Creator and maintainer of an open-source library providing interactive 3D components for React, built natively on Three.js, React Three Fiber (R3F), and GSAP.
- Engineered modular, production-ready assets allowing developers to seamlessly drop scroll-reactive surfaces, particle fields, instanced layouts, and post-processing into any React site or web application.
- Implemented strict clean code architecture and mathematical optimizations to ensure the package is lightweight, highly performant, and developer-friendly.

Reddit Sentiment Analysis (NLP)

- Engineered a deep NLP pipeline using fine-tuned BERT for nuanced emotion classification on large-scale Reddit data.
- Practiced advanced prompt engineering and integrated LLM-based interfaces with frontend applications.

EDUCATION

BSc. Computer Science and Information Technology (CSIT) 2022 – April 2026
Prime College, Tribhuvan University | Kathmandu, Nepal

REFERENCES

Available upon request.